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MIL-M-23167B(SHIPS)
29 December 1969
SUPERSEDING
MIL-M-23167A(SHIPS)
10 April 1967
(See 6.7)

MILITARY SPECIFICATION

METER, FREQUENCY, EXPANDED SCALE SWITCHBOARD TYPE

(NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope. This specification covers high accuracy, shockproof switchboard mounted, expanded scale frequency meters for Naval shipboard use.

1.2 Classification.

1.2.1 Type designation. The type designation of frequency meters shall be in the following form and as specified (see table IV and 6.2):

_MR_	_49_	_W_	_41_	_AF_	_H_
Component (1.2.1.1)	Style (1.2.1.2)	Color scheme (1.2.1.3)	Center scale value and voltage rating (1.2.1.4)	Kind of current 1.2.1.5)	Special feature (1.2.1.6)

1.2.1.1 Component. Frequency meters covered by this specification shall be designated by the two letter symbol "MR".

1.2.1.2 Style. The style shall be designated by a two digit number which signifies the characteristics specified in 1.2.1.2.1 and 1.2.1.2.2.

1.2.1.2.1 First digit. The first digit shall designate the normal size and shape of flange as follows:

3 – 3-1/2 inch round flange, sealed.
4 – 4-1/2 inch rectangular flange.

1.2.1.2.2 Second digit. The second digit shall designate the type mounting and the normal scale length in angular degrees as follows:

8 – Flush mounting, 90 degree nominal scale.
9 – Flush mounting, 250 degree nominal scale.

1.2.1.3 Color scheme. The color scheme of the dial background, markings, and pointer shall be designated by a single letter as follows:

W – White dial background, black markings and black pointer.

1.2.1.4 Center scale value and nominal voltage rating. The center scale value and nominal voltage rating shall be designated by two digit numbers as specified in 1.2.1.4.1 and 1.2.1.4.2.

1.2.1.4.1 First digit. The first digit shall designate the center-scale value as follows:

4 – 400 Hertz: (Hz) scale range 395-405 Hz

1.2.1.4.2 Second digit. The second digit shall designate the nominal voltage rating as follows:

1 – 120 volts.
2 – 450 volts.

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3 – 208 volts.
4 – 220 volts.
5 – 440 volts.

1.2.1.5 Kind of current. Two letters shall designate the kind of current for which the meter is designed as follows:

AF – Alternating current (a.c.) 400 Hz nominal frequency.

1.2.1.6 Special feature. A single letter shall designate the special feature as follows:

H – High-impact-shock resistant.

1.2.1.7 Example of type designation. The type designation MR38W41AFH signifies:

(a) MR – Frequency meter.
(b) 38 – 3-1/2-inch round flange sealed, flush mounted 90-degree nominal scale.
(c) W – White dial background with black markings and black pointer.
(d) 41 – Center-scale value 400 Hz, 120 volt nominal voltage rating.
(e) AF – Alternating current 400 Hz, nominal frequency.
(f) H – High-impact, shock resistant.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-14	– Molded Plastics and Molded Plastic Parts, Thermosetting.
MIL-S-901	– Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	– Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-M-10304	– Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-E-15090	– Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-M-16034	– Meters, Electrical-Indicating (Switchboard and Portable Types).

STANDARDS

MILITARY

MIL-STD-130	– Identification Marking of U.S. Military Property.
MIL-STD-167	– Mechanical Vibrations of Shipboard Equipment
MIL-STD-761	– Electric Power, Alternating Current, for Shipboard Use, Characterization and Utilization of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)
C39. 1-1964 – Electrical Indicating Instruments, Requirements for.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N.Y. 10016.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Qualification. Frequency meters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Material. The material for each part shall be as specified herein. When a definite material is not specified, a material shall be used which is adequate for the purpose intended and for meeting the requirements specified herein. Approval or acceptance of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Molded thermosetting plastic material. Molded thermosetting plastic material conforming to MIL-M-14 shall be used.

3.2.2. Encapsulating material. Encapsulating material where used, shall be in accordance with MIL-E-917.

3.3 Design and construction of parts and subassemblies.

3.3.1 Indicators.

3.3.1.1 Three and one-half inch size. The 3-1/2 inch size indicators shall conform to MIL-M-10304.

3.3.1.2 Four and one-half inch size. The 4-1/2 inch size indicators shall conform to MIL-M-16034.

3.3.2 Cases. Cases shall be made of molded thermosetting plastic material or metal. Metal cases shall be protected against corrosion in accordance with MIL-E-917.

3.3.2.1 Finish. The portion of the case exposed to view from the front of the panel shall have a light gray finish in accordance with formula No. 111, class 2 of MIL-E-15090. No nickel or other bright trimmings shall be used. Metal cases shall be rendered resistant to corrosion prior to the application of the final finish (see 3.3.2).

3.3.3 Expanded scale circuit. The expanded scale circuit parts shall be encapsulated with varnish or compound (see 3.2.2) for environmental protection.

3.3.4 Scale lengths. Scale lengths of instruments shall be as follows:

(a) 3-1/2 inch round 90-degree scale indicator – 2-1/2 inches minimum.
(b) 4-1/2 inch rectangular 250-degree scale indicator – 5 inches minimum.

3.3.5 Scale divisions. Scale span in both sizes shall be 395-405 Hz. Scale divisions shall be as follows:

(a) 3-1/2 inch round 90-degree scale indicator shall be marked so that each scale division represents 0.2 Hz.
(b) 4-1/2 inch rectangular 250-degree scale indicator shall be marked so that each scale division represents 0.1 Hz.

3.3.5.1 The color scheme for the dial shall be white dial background, black markings and black pointer.

3.3.6 Mechanical zero point. The mechanical zero shall be identified on the scale. The division mark corresponding to the zero point shall be marked in red and identified by the term “Zero-Adjust” also marked in red.

3.3.7 Zero-adjuster. Frequency maters shall have a zero-adjuster accessible from the front of the case. It shall provide for a range of adjustment above and below the zero point of not less than 3 percent of scale length and preferably be capable of rotating through 360 degrees without damage to any part of the meter and without becoming inoperative. If the zero-adjuster is not designed for 360-degree rotation, it shall be capable of withstanding a torque of 10 ounce-inches without damage to the zero-adjuster or any part of the meter. The zero-adjuster shall have sufficient friction to prevent it from shaking loose and changing adjustment when subjected to tests specified herein.

3.3.8 Terminals. Terminal studs shall be firmly anchored to the base of the meter in such a manner as to prevent turning or movement sufficient to cause damage to internal construction. Each terminal stud shall be furnished with the necessary nuts or screws, and washers (see figures 1 and 2).

3.3.9 Self-contained limitations. Frequency meters furnished under this specification shall be self-contained.

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3.3.10 Mounting. Frequency meters shall be mounted on a panel drilled in accordance with the applicable drilling plan, and shall be mounted by means of mounting bolts or studs (see figures 1 and 2). The necessary mounting bolts (screws), nuts and lockwashers shall be supplied with each meter.

3.4 Calibration.

3.4.1 Position. Frequency meters shall be calibrated in the position in which they are normally used, and shall meet the accuracy requirements specified herein in such position at the reference temperature. Normal position is defined as with the scale in a vertical plane.

3.4.2 Mounting panel. The Frequency Meters shall meet the accuracy requirements specified herein when mounted on a magnetic steel panel 1/8 inch thick.

3.5 Initial accuracy. The initial accuracy expressed as a percentage of center-scale and in absolute value shall be within the limits specified in 3.19 (see 4.3.4).

3.6 Harmonic content effect. The frequency meter shall not exceed the error specified in 3.19 when subjected to the test specified in 4.3.16.

3.7 Voltage range. Frequency meters shall operate over the range specified in 3.19 and the change in indication from the reading at nominal voltage shall not exceed the value specified (see 4.3.7).

3.8 On-off cycling. Meters shall withstand on-off cycling test specified in 4.3.8 without change in indication exceeding that specified in 3.19.

3.9 Overvoltage. Meters shall withstand the test of 4.3.9 without damage or permanent change in indication exceeding the value specified in 3.19.

3.10 External field influence. Frequency meters shall have sufficient magnetic shielding so that the effect of an external field of 5 oersteds will not cause a change in indication greater than the value specified in 3.19 (see 4.3.10).

3.11 Temperature range. Frequency meters shall indicate freely (see 6.5.3) during and after the test specified in 4.3.11. The permanent change in indication shall not exceed the values specified in 3.19.

3.12 Inclination. Frequency meters shall indicate freely (see 6.5.3) during the test specified in 4.3.6. The change in indication shall not exceed the value specified in 3.19.

3.13 Vibration. When tested in accordance with 4.3.14 no screws, bearings, pivots or other parts shall be so loosened or damaged as to cause a change in indication greater than the value specified in 3.19.

3.14 Shock (high-impact). When tested in accordance with 4.3.15 no screws, bearings, pivots, windows or other parts shall become loosened, cracked or unduly damaged. The permanent change in indication based on calibration made after resetting the pointer to zero shall not exceed the value specified in 3.19.

3.15 Dielectric stress. When tested in accordance with 4.3.12, frequency meters shall withstand a voltage of 2,600 volts a.c. without damage or flashover.

3.16 Insulation resistance. The insulation resistance of frequency meters shall be not less than 20 megohms (see 4.3.13).

3.17 Marking. Frequency meters shall be permanently and legibly marked with the applicable information specified in 3.17.1.

3.17.1 Dials. Dial markings shall be sharply defined and visible from the front of the case. These markings shall not however, be so conspicuous as to distract attention from the scale markings. The scale markings and numerals and items (a), (b), (c), and (d) shall be marked on the dial; items (a) through (h) shall be marked on the dial, on the front of the meter, or on an attached identification plate. Marking shall conform to MIL-STD-130.

(a) Designation of the quantity measured (Hertz).
(b) Nominal voltage rating (120, 450 and so forth as applicable).
(c) Type designation (see 1.2.1).
(d) MIL-M-23167.

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(e) Manufacturer's name or trade mark (or symbol), or both.
(f) Manufacturer's model number.
(g) Manufacturer's serial number.
(h) Federal stock number.

3.18 Burden. Design of expanded scale circuit shall be such as to require a minimum amount of energy to energize the meter. Burden shall not exceed 7.5 volt-amperes.

3.19 Performance. Basic initial accuracy of meters shall not exceed 0.05 percent (0.2 Hz) within the scale span of 398 to 402 Hz and 0.1 percent (0.4 Hz) at all other scale points. Permissible errors are listed in table I in absolute values for the respective reference conditions.

Table I – Permissible errors, Hertz

Condition	Scale spans	
	398 – 402 Hz	395 – 398 402 – 405 Hz
Initial accuracy	0.2	0.4
Voltage range	.4	.8
On-off cycling	.2	.4
Overvoltage	.2	.4
External field influence	.2	.4
Inclination	.2	.4
Temperature range	.4	.8
Harmonic content:		
3% single harmonic	.1	.2
4% single harmonic	.2	.4
5% single harmonic	.3	.6
Vibration	.4	.8
Shock (high-impact)	.6	1.2

3.20 Workmanship. Meters shall be manufactured in a careful manner in accordance with good design and sound practice. The interiors of meters shall be free from metal filings, grease or oil, foreign material, dust or other loose particles which may interfere with normal operation of the meter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The methods of sampling, examination and tests conducted on frequency meters shall be classified as follows:

(a) Qualification (see 4.3).
(b) Quality conformance (see 4.4).

4.2 Test conditions.

4.2.1 Atmospheric conditions for qualification tests. Unless otherwise specified herein, qualification tests shall be made at ambient temperatures of 23° to 27° Centigrade (C.) at existing barometric pressure, and with relative humidity of between 40 to 60 percent.

4.2.2 Quality conformance tests. Unless otherwise specified herein, all quality conformance tests shall be made at ambient temperature of 20° to 30°C. with a relative humidity of less than 80 percent.

4.2.3 Position. Unless otherwise specified herein, tests shall be conducted with the frequency meter in the position in which it is normally used.

4.2.4 Tapping. When taking readings, either the meter or its support shall be tapped lightly, except for ascertaining the effect of friction after the vibration test.

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4.2.5 Parallax. Care should be taken to avoid the effect of parallax when taking meter readings.

4.2.6 Resetting to zero. Unless otherwise specified herein, meters may be set to zero with the use of the zero-adjuster at the completion of each test.

4.2.7 Thermal stability. For all tests, care shall be taken to obtain thermal stability at the temperature specified herein prior to reading the meter.

4.2.8 Number of scale reading. Unless otherwise specified herein, readings shall be taken at each major division between 398 Hz and 402 Hz. Two additional readings shall be taken at the extreme ends of the scale.

4.2.9 Test equipment. Standard laboratory instruments shall be used to check the accuracy of indications of frequency meters. Approval of the laboratory instruments, by the Naval Ship Engineering Center (NAVSEC), shall be required prior to their use in checking the accuracy of the meters covered by this specification.

4.3 Qualification tests.[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in Table II.

4.3.1 Samples. Samples submitted for qualification may be either representative of the manufacturer's normal production or a custom made sample. Two meters of each size (3-1/2 or 4-1/2 inch) for which qualification is desired shall be subjected to the examination and tests specified in table II. Examination and tests shall be conducted in the order listed. Meters may be of any voltage rating.

Table II – Qualification examination and tests

Examination and test	Requirement paragraph	Examination and test paragraph
Visual and mechanical examination (without opening case)	---	4.3.2
Zero-adjuster	3.3.7	4.3.3
Burden	3.18	4.3.5
Initial accuracy	3.5	4.3.4
Harmonic content	3.6	4.3.16
Voltage range	3.7	4.3.7
On-off cycling	3.8	4.3.8
Overvoltage	3.9	4.3.9
External field influence	3.10	4.3.10
Inclination	3.12	4.3.6
Temperature range	3.11	4.3.11
Dielectric stress	3.15	4.3.12
Insulation resistance	3.16	4.3.13
Vibration	3.13	4.3.14
Shock (high-impact)	3.14	4.3.15

4.3.2 Visual and mechanical examination. Frequency meters shall be examined to determine conformance with table III.

Table III – Visual and mechanical examination

Requirement	Paragraph
Material	3.2
Design and construction	3.3
Mounting	3.3.10
Marking	3.17 and 3.17.1
Workmanship	3.20

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

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4.3.3 Zero adjustment. The range of zero adjustment above and below the center of the zero point shall be determined. Record shall be made as to whether or not the zero-adjuster may be rotated through 360 degrees without damage to the meter in addition to whether the zero-adjuster is designed for 360-degree rotation and is capable of withstanding a torque of 10 ounce-inches without damage to the meter.

4.3.4 Initial accuracy. Frequency meters shall be conditioned in the test circuit until thermal stability is attained. The meter shall be energized at the center-scale value. Reading shall be taken and compared to the reading of the laboratory standard instrument. The test shall be made with the meter at a temperature of 45° ± 2°C.

4.3.5 Burden. The burden shall be measured at center-scale deflection and recorded in terms of volt-amperes to determine conformance with 3.18.

4.3.6 Inclination. Frequency meters shall be mounted in the inclined positions as follows and check for accuracy made at each position as specified hereinafter:

(a) 60 degrees forward of normal.
(b) 60 degrees backward of normal.
(c) 60 degrees to right of normal.
(d) 60 degrees to left of normal.

4.3.7 Voltage range. Frequency meters shall be checked for accuracy at voltages specified in table IV, as applicable.

Table IV – Applied voltages at which meters are to be tested.

Voltage rating of meter	Type designations	Test voltage	
120	MR38W41AFH	114	126
120	MR49W41AFH	114	126
450	MR38W42AFH	428	472
450	MR49W42AFH	428	472
208	MR38W43AFH	199	218
208	MR49W43AFH	199	218
220	MR38W44AFH	209	231
220	MR49W44AFH	209	231
440	MR38W45AFH	418	462
440	MR49W45AFH	418	462

4.3.8 On-off cycling. Meters shall be subjected to a test of 2000 cycles of application and removal of rated voltage. The repetition rate of these cycles shall be sufficiently low so that the meter indicator momentarily comes to rest upon both application and removal of voltage. Lower repetition rates may be used in those cases where this test may cause excessive heating of the meter movement or scale expansion circuit. Change in indication as a result of this test shall be determined (see 3.8).

4.3.9 Overvoltage. Meters shall be subjected to 135 percent rated voltage for 15 minutes. Change in indication as a result of this test shall be determined (see 3.9).

4.3.10 External field influence. Frequency meters shall be placed at the center of a four turn coil 50 centimeters in diameter, carrying a current of 50 amperes of the same kind and frequency as that which actuates the meter mechanism and which produces a field of approximately 5 oersteds at the center of the coil. Readings shall be taken with the meter at various positions in the center of the field (see table I).

Note. – It is the intent to specify a test coil which will produce a uniform field of approximately 5 oersteds through the case of the meter. Therefore, coils, of other sizes may be used if an equivalent magnetic field is produced.

4.3.11 Temperature range. Frequency meters shall be maintained at each extreme temperature (0° and 65° C) for a sufficient time to attain thermal stability (approximately 2 hours). At the end of this period, readings shall be taken at each temperature. The effect of temperature shall not exceed the values specified (see 3.11).

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4.3.12 Dielectric stress. When performing this test, meters with nonmetallic cases shall be mounted on a metallic panel. All meters shall be subjected to a voltage of 2,600 volts r.m.s. applied between all external live parts of the meter and:

(a) The metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic.
(b) The zero-adjuster; and range selector switch, when applicable.
(c) All exposed metal parts not a live part of the meter.

4.3.12.1 The voltage shall be nominal 60 Hz in frequency and shall approximate, as closely as possible, a true sine wave in form. The voltage shall be applied continuously for a period of 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer or by means of an auxiliary ratio transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually, and upon completion of the 60-second test period shall be gradually reduced. The test shall be made using high-potential test prods having not less than 1/8 inch radius tips. There shall be no signs of damage or flashover.

4.3.13 Insulation resistance. Meters shall be placed on a metallic platform or mounted on a metallic panel as specified in 4.3.12. Using a potential of 500 volts direct current, the insulation resistance shall be determined between all external live parts and the metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic (see 3.16).

4.3.14 Vibration. Frequency meters shall be subjected to vibration in accordance with type I of MIL-STD-167.

4.3.15 Shock (high-impact). Frequency meters shall be subjected to high-impact shock tests in accordance with type C of MIL-S-901 for lightweight equipment when mounted on a plate as shown on fixture 6D-1 as shown on figure 7-1 of MIL-S-901. Both test samples shall be subjected to the test. During the test, one frequency meter shall be energized to the center-scale value. The pointer may be reset to zero after each blow. At the completion of the test, the meters shall be checked for calibration errors (see table I).

4.3.16 Harmonic content. Voltage containing harmonic-frequency energy shall be applied to the meters as follows:

(a) Fundamental (of nominal voltage rating) plus 3 percent each harmonic, 2nd through 20th, in turn.
(b) Same as (a), except 4 percent harmonic content.
(c) Same as (a), except 5 percent harmonic content.

4.4 Quality conformance inspection. Each meter shall be subjected to the examination and tests specified in table V. Examination and tests shall be conducted in the order listed. Frequency meters which do not meet requirements specified herein shall be rejected.

Table V – Quality conformance inspection.

Examination and tests	Requirement paragraph	Examination and test paragraph
Visual and mechanical examination (without opening case)	---	4.3.2
Initial accuracy	3.5	4.3.4
Temperature range	3.11	4.3.11
Voltage range	3.7	4.3.7
Insulation resistance	3.16	4.3.13
Dielectric stress	3.15	4.3.12

4.5 Inspection of preparation for delivery. Packaging, packing and marking of packages shall be inspected to determine conformance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

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5.1 Domestic shipment and early installation.

5.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules, or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include type designation, Federal stock number or manufacturer's serial or model number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.2 Preservation, packaging, and packing. The equipment shall be preserved and packaged by levels A or C and packed by levels A or B in accordance with MIL-E-17555.

5.2.3 Marking. In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use. Frequency meters furnished under this specification are intended for use on type III, 400 Hz power systems of MIL-STD-761.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Complete type designation, (see 1.2.1).
(c) Federal stock number.
(d) Preservation, packaging, packing or marking requirements if other than those required by 5.1 (see 5.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-23167 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Definitions. Definitions of terms used in this specification shall be in accordance with USAS C39.1, except as otherwise specified herein.

6.5.1 Accuracy. The accuracy shall be a number which defines the limit of error, expressed as a percentage of center-scale value.

6.5.2 Accuracy, rated. The rated accuracy is a number assigned by the manufacturer to designate the accuracy classification of the frequency meter. This number, usually expressed as a percentage of center-scale value is given as the limit which errors will not exceed when the meter is used under reference conditions. This specification, however, lists permissible errors in absolute values.

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6.5.3 Free indication. Meters shall be considered as indicating freely if the pointer of the meter under test moves simultaneously with the pointer of the standard meter with which it is being compared when the frequency of the electrical energy supplied to both meters is gradually changed.

6.6 Precautions. Attention of the equipment designers and manufacturers is invited to the fact that sufficient space should be provided in equipment for mounting standard meters of the maximum depth specified (see figures 1 and 2) even though a shallower meter may be specified in the initial design.

6.7 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENEINCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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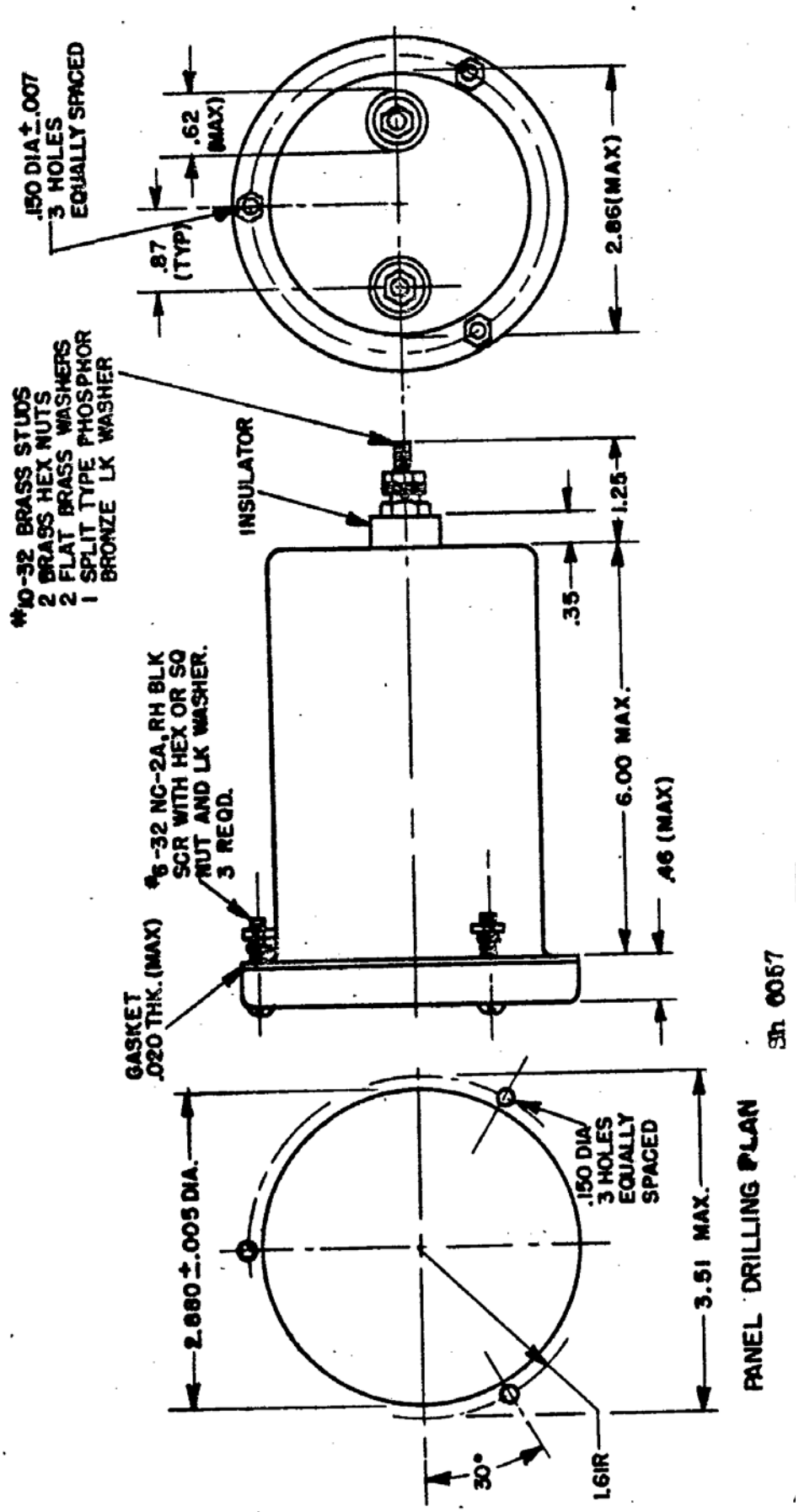


Figure 1. Size and mounting dimensions, MR38 series

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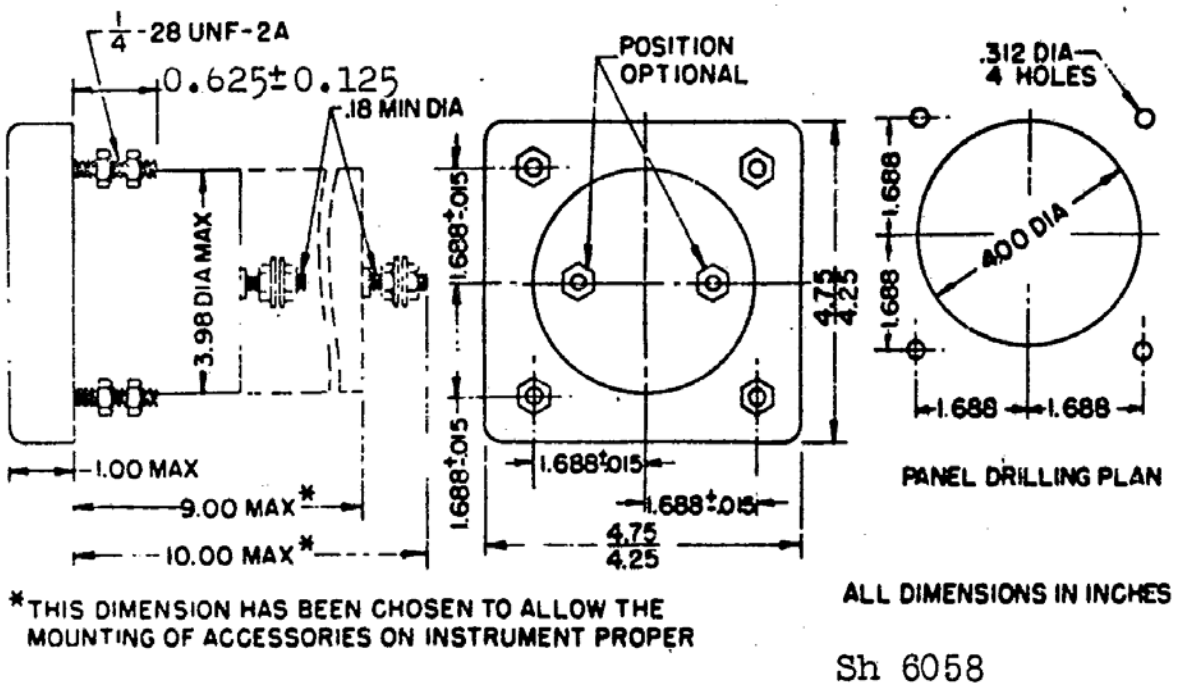


Figure 2. Size and mounting dimensions, MR49 series

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